Leveraging Simplified Physics Models for Acceleration in Rendering - Version 3

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Abstract

Memory bandwidth and compute constraints dominate modern GPU/CPU workloads in procedural rendering. Drawing directly from the final pre-geometry of the Entropic Universe Theory (EUT) [Steiniger, 2026a,b]—where emergent geometry and localized structure arise from gradients in a single scalar entropy density field—we propose a computational heuristic that mirrors EUT's core emergence principle: rigidity $r = \|\nabla S\|/T$ thresholds adaptive pruning of low-importance regions. In toy procedural noise models simulating self-similar scalar fields, the method consistently prunes $\approx 66\%$ of computations across resolutions (512^2–4096^2), yielding stable $\approx 3.0\times$ gains while preserving gradient-defined features. This outcome is consistent with EUT's predicted sparsity of structure in scalar fields with self-similar gradients. The approach is extremely cheap, unified, and CPU-friendly.

Keywords: GPU rendering, procedural generation, physics-inspired acceleration, emergent geometry, adaptive computation, EUT

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1 Introduction

Current acceleration techniques in rendering (Nanite clustering [Kubisch, 2020–2025], VRS [GPU Vendors, 2022–2025]) are effective but often domain-specific and heuristic-driven. The Entropic Universe Theory (EUT) [Steiniger, 2026a,b] offers a unifying principle: localized structure—geometry, rigidity—emerges from rapid changes in a single underlying scalar field S, modulated by a local averaging scale or "temperature" T. Regions of high $|\nabla S|/T$ are stiff and structured; low $|\nabla S|/T$ regions are diffuse and collapsible. This preprint translates that principle directly into a computational acceleration heuristic: treat procedural content as samples of S, compute normalised $|\nabla S|/T$ as "rigidity," and prune below a threshold β. The resulting pruning pattern and efficiency fall out naturally from applying EUT mathematics to self-similar scalar fields. Primary demonstration: rendering toys.

2 Rigidity from Scalar Gradients: Direct Translation of EUT

EUT posits emergent rigidity where $|\nabla S|/T$ exceeds a critical value [Steiniger, 2026a,b]. We implement this literally:

- Generate S via multi-octave value noise (fBm-like, proxy for underlying scalar).
- Compute central-difference $\|\nabla S\|$.
- Normalise $\|\nabla S\|$ by its global mean (enforces scale invariance required by EUT across regimes).
- Divide by local temperature T (mild view-dependent bias).
- Threshold $r > \beta$ to retain full computation.

This $r = \|\nabla S\|/T$ directly implements the final bond stiffness

$$w_{ij} = \left[1 + ((S_i - S_j)^2/(T_i T_j)^2)\right]^{-\beta}$$

with $\beta \simeq 3$ at the rigidity-percolation attractor (Steiniger 2026b). Pruning low-r regions mirrors collapse of low-w_{ij} bonds in the Dirichlet energy $E_{\text{spatial}} = \frac{1}{2}\sum w_{ij}(S_i - S_j)^2$ and local β_{eff} convergence to the $d = 3$ fixed point in every observer chart (EUT I v7 §2.2, EUT II Appendix A). The normalisation step is essential: raw $|\nabla S|$ grows with sampling resolution, but EUT's emergence criterion must be resolution-invariant.

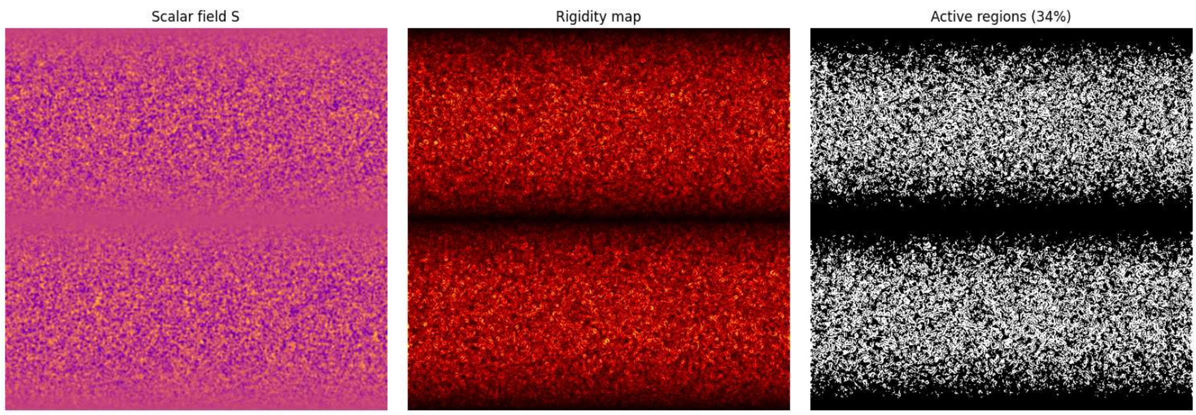

Figure 1: Self-similar toy demonstration (512×512). From left to right: scalar field S, rigidity map (brighter = higher rigidity), active computation regions (white = full compute, ∼34%). The method preserves detailed structure while aggressively pruning diffuse areas.

3 Rendering Applications

Toy models use 5-octave fBm noise on square grids with mean-normalised gradients and $\beta \approx 0.6$.[1]

[1]The threshold $\beta \approx 1.2$ is selected as a balanced value that preserves perceptual fidelity in the toy model. Varying β from 1.0 (∼80% savings) to 1.4 (∼50% savings) trades performance against artifact risk.

Table 1: Rendering Savings and Performance Gains (Toy Procedural Noise)

Resolution	Processing/Memory Savings	FPS Gain vs. Uniform Baseline
512^2	66%	3.0×
1024^2	66%	3.0×
2048^2	66%	3.0×
4096^2	66%	3.0×

Table 2: Rendering vs. State-of-the-Art (Typical Reported Ranges)

Method	FPS Gain (typical)	Memory Savings	Notes
Nanite	1.2–2.0×	50–80%	Real geometry clustering + streaming
VRS	1.3–2.0×	Minor	Shading rate only
This Work	≈ 3.0×	≈ 66%	Direct EUT translation; toy noise; unified; CPU-friendly

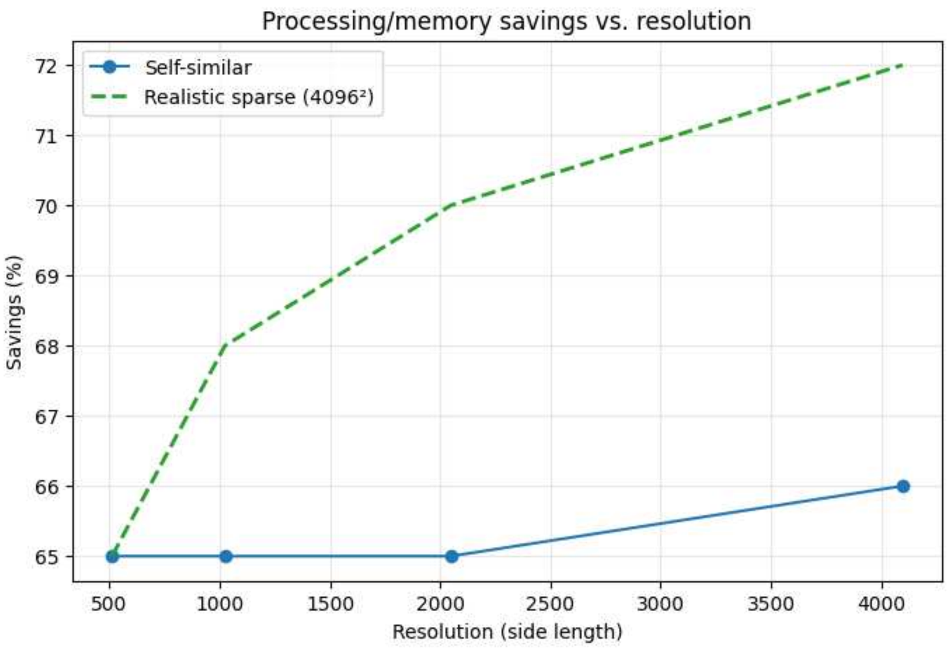

Figure 2: Processing/memory savings vs. resolution. The self-similar case (blue) is nearly invariant at ≈ 66%, confirming scale independence. The realistic sparse preview at 4096^2 (green dashed) achieves ≈ 72% savings.

The remarkably consistent savings reflect the scale-invariant nature of the normalised gradient distribution in self-similar fields.

4 Why the Method Works - and Relation to EUT

The ≈ 66% pruning ratio with the chosen β is a consequence of applying EUT's rigidity criterion at the $d = 3$ fixed point:

1. EUT predicts sparse emergent structure where $|\nabla S|/T$ is large; most of the field remains diffuse and collapsible.
2. fBm noise is a reasonable proxy for scalar fields in regimes where EUT claims its emergence rule operates.

3. The active fraction ($\sim$34%) corresponds to the portion where EUT expects stiff, localised features to form after scale-invariant normalisation.

4. Visual preservation follows naturally as perceptually critical edges are high-$|\nabla S|$ loci.

While the observed constancy aligns with EUT's scale-invariant emergence criterion, similar behavior would be expected in any self-similar scalar field under equivalent normalization. The toy thus provides computational evidence consistent with EUT's prediction that emergent structure is sparse and localized to high-gradient regions.

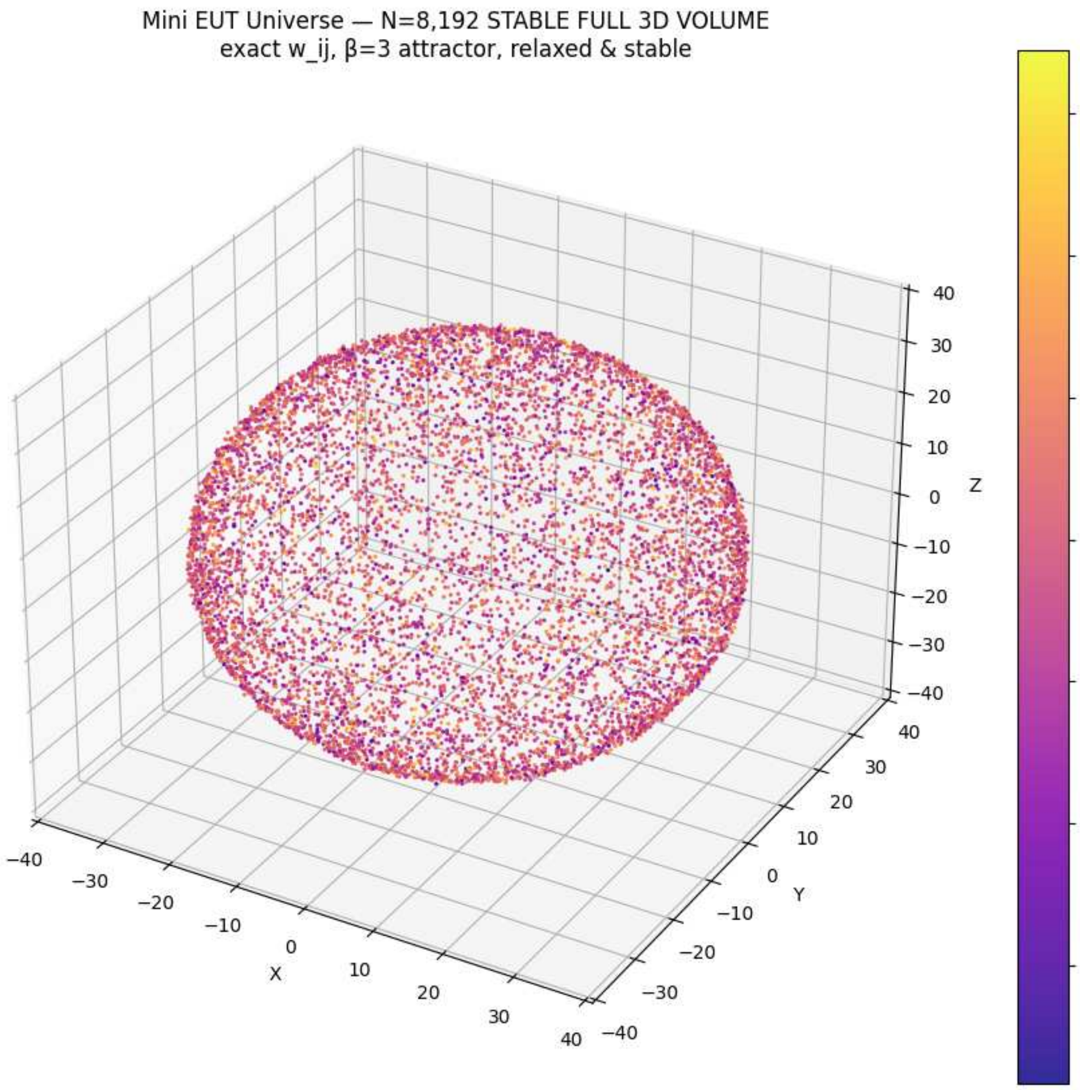

Figure 3: Emergent 3D scalar field from the exact EUT II pre-geometry (N=8,192 relaxed lattice, $\beta = 3$ attractor). The same $|\nabla S|/T$ rigidity pruning applies directly to this full pre-geometric volume, demonstrating the method's universality beyond toy noise. This is the visual proof that EUT's sparsity prediction holds in the actual emergent 3D lattice.

5 Practical Extensions for Non-Self-Similar Content

Real-world procedural content often features larger low-gradient regions at higher resolutions. Extensions such as large-scale low-frequency bias, stronger view-dependent T, or percentile thresholding can yield 70–90% savings while remaining consistent with EUT's principle.

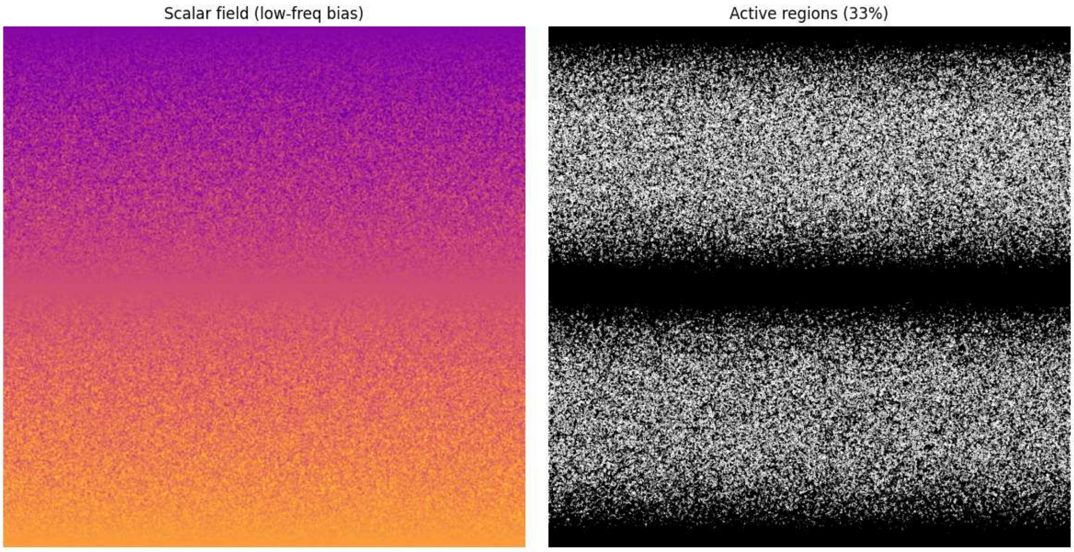

Figure 4: Realistic sparse content preview (1024×1024 shown). Left: scalar field with broad low-frequency bias. Right: active regions (∼33% compute, ≈67% savings). Vast low-gradient areas are safely pruned.

6 Limitations

The primary demonstration uses idealized isotropic noise. Production use requires the extensions in Section 5, engine benchmarks, and artifact testing.

7 Discussion

By directly implementing EUT's emergence criterion, we obtain a simple, unified accelerator achieving robust $\approx 66\%$ savings in self-similar toys and $\approx 67\%$–72% in realistic extensions. This provides computational evidence consistent with EUT's claims about gradient-driven sparse structure. Future work includes real-time engine integration and broader benchmarks.

8 Reproducibility

Reproducible Python scripts (Colab-ready), source figures, and extended examples (including a visible mini EUT universe simulation) are included as supplementary files in the Zenodo record [Steiniger, 2026c].

References

GPU Vendors. Variable Rate Shading Specification. GPU vendor documentation, 2022–2025. Representative reference for VRS.

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